



CN Complete

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1-08-24

Exploring the OSI model and TCP/IP

OSI - 7 layers } Two models
TCP/IP - 5 layers }

OSI - Open System Interconnection
TCP - Transmission Control Protocol

→ The need of networking model.

- Network: A network is defined as having two or more computing devices interconnected, using a set of rules (protocols) that allows them to share resources between themselves.

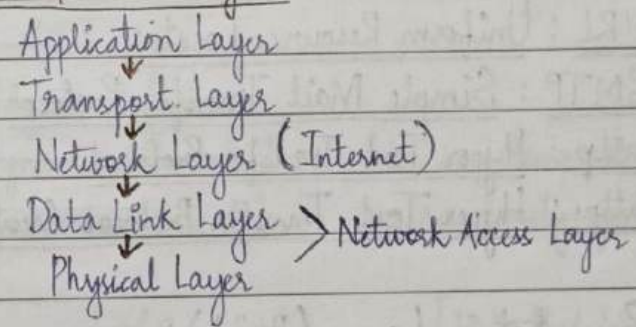
Organizations provide role in Internet -

- ① ISOC - Information Security Operations Center
- ② IETF - Internet Engineering Task Force

OSI - 7 layers



TCP/IP - 5 layers



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OSI Model :

- OSI model is developed to be an open networking model for computer network to allow different devices to use a set of mutual protocol to allow communication between each other over a network.
- At each layer of OSI model when a message exists at a specific layer, the message is called as PDU (Protocol Data Unit).
- As the PDU is created at the Application Layer, it is referred as data / raw message / datagram.
- As the PDU travels down the OSI model, each of the lower layer is responsible for attaching additional information within a header onto the PDU. This process is called as Encapsulation.
- When a host on the network receives the message, the PDU travels upwards on the OSI model, where each layer de-encapsulate the message, removing

the header information until the raw message is delivered to the Application Layer on the receiver's side.

- Application Layer (AL), Presentation Layer & Session Layer (SL) are designed to ensure the datagram that is created by the sender can be transmitted across the network between the sender & the receiver.
- Transport Layer (TL), Network Layer (NL), Data Link Layer (DLL), Physical Layer (PL) they focus on inserting the addressing information that are needed to deliver the datagram to the destination.

Application Layer (AL):

- Provides an interface so that we can run the application of a host (comp/smart phone) to communicate with the protocol of the OSI model.
- URL: Uniform Resource Locator. • FTP: File Transfer Protocol
- SMTP: Simple Mail Transfer Protocol. • DNS: Domain Name System
- http: Hyper Text Transfer Protocol
- https: Hyper Text Transfer Protocol Secure.

Presentation Layer (PL):

- Function.
- AL protocol creates system dependent data (ASCII/JPEG) whereas Presentation Layer transforms it into an independent format.
 - Responsibility of Presentation Layer -
 - Data formatting
 - Data compression
 - Data encryption
 - Data decryption

Session Layer (SL):

- Create/establish a session.
- Maintain the session.
- Terminate the session.

Transport Layer (TL):

- The purpose of TL is to assign a service port number to the PDU so that the receiving system will know how the presentation layer will interpret and format the data. Then the receiving system can read the data in Application Layer (AL).

- It assigns a service port number to the PDU.

Port range	Category
① 0-1023	Well known port (belonging to AL protocol)
FTP	20 & 21
SMTP	25
DNS	53
HTTP	80
HTTPS	443
② 1024-49,151	Registered Port
③ 49,152-65,535	Dynamic/Private Port which are temporarily used for communication & are randomly generated.
(Ephemeral Port Number)	

Source Port Number	Data	→ (Segment)	[PN: Port Number]
Destination Port Number			

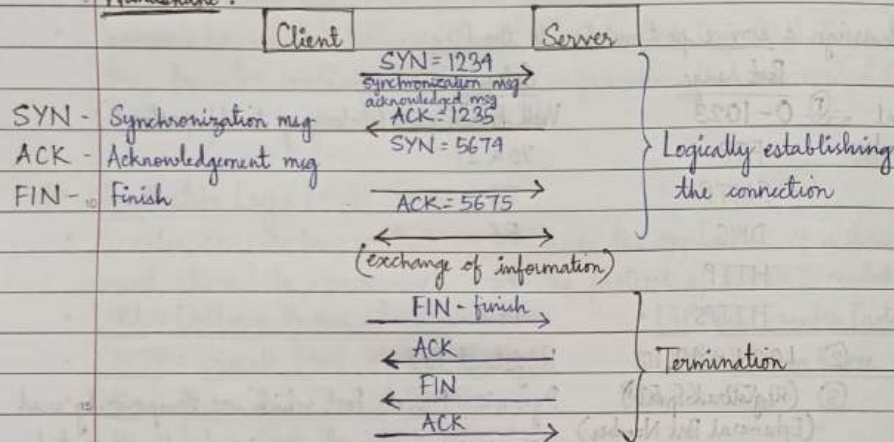
Client → Server	Server → Client
Source IP address: 192.168.1.10	Source IP address: 192.168.1.100
Destination IP address: 192.168.1.100	Destination IP address: 192.168.1.10
Source Port Number: 49161	Source Port Number: 80
Destination Port Number: 80	Destination Port Number: 49161

HTTP Request Message

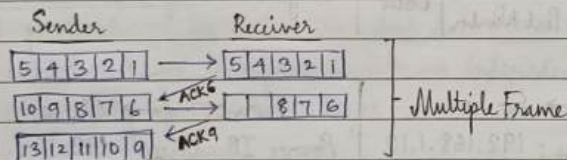
HTTP Response Message

- Two types of protocols on Transport Layer:
- ① TCP: Transmission Control Protocol
 - ② UDP: User Datagram Protocol

- ① TCP (Transmission Control Protocol) - Connection Oriented Protocol → reliable connection.
- This protocol is connection based protocol / connection oriented protocol that establishes a logical connection between the source and the destination device before exchanging message over a network. This connection is referred as Three Way Handshake.



- Connection Oriented Protocol is also called as reliable connection.



05-09-24 ⇒ Advantage of TCP -

- It establishes a session before exchanging information.
- Provide reliability in delivering the data over the network.
- Deliver the data in same order of delivery from the sender to the receiver.
- It provides flow control by creating a window size which is been mutually agreed between sender & the receiver.

⇒ Disadvantage of TCP -

- It has a lot of overhead.
- When a host is sending multiple messages to another host over a network, the

messages are not sent at a time. TCP creates a logical window that specifies how much data can be sent at a time. TCP will send several messages within the window & wait for an acknowledgement (sender is remaining idle).

② UDP (User Datagram Protocol) -

- It is connectionless, unreliable, doesn't wait for any acknowledgement, doesn't provide or use any sequence number.

⇒ Advantages of UDP -

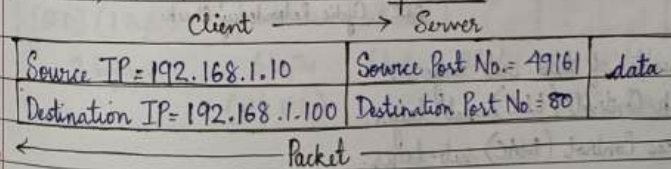
- Since UDP doesn't wait for acknowledgement, the messages are being sent faster over the network.
- It has low / less overhead.

⇒ Disadvantages of UDP -

- It has no error detection.

Network Layer (NL) : / Internet Layer

- It provides logical addressing information into the datagram.
- It provides a unique IP address to a device present inside a network.
- The address are provided according to IP version 4 (or) IP version 6.
- In NL, the PDU is called as packets.



- It is responsible for routing services across the network.

Router is a physical device consists of 3 layers.

1. Physical Layer
2. Data-Link Layer
3. Network Layer

- Protocol used in NL is Internet Protocol (IP).

⇒ Internet Protocol :

Connectionless Protocol that does not establish any logical connection between sender and the receiver. Therefore, if packets are lost / corrupted during the

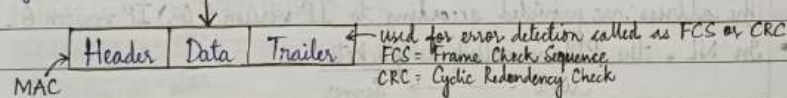
- It is unreliable and doesn't provide any guarantee that the data will be delivered to the destination host.
- Internet Protocol (IP) provides less overheads.

⇒ MTU (Maximum Transmission Unit):

- It describes the maximum size of a message that can be supported by network media and the default size is 1500 bytes. (wired, wireless, fiber optics)
- Network Media / Transmission Media / Channel / Path / Line: Wired (co-axial cable, twisted pair cable, etc.), Wireless, Fiber Optics.

Data-Link Layer (DLL):

- The purpose of DLL is to be responsible for moving the datagram from the upper-layer onto the actual network.
- It handles the flow control onto the media and also manages the flow control of incoming messages from the physical layer to the upper layer.
- The DLL also access the network media.
- The PDU in DLL is called as frames.
- It handles error detection to identify whether an incoming frame from the physical network are corrupted.



- The DLL consists of two sub-layers:
 - i) Logical Link Control (LLC) sub-layer.
 - ii) Media Access Control (MAC) sub-layer.

i) Logical Link Control (LLC):

- It is responsible for ensuring communication between the networking application, software, protocol of the upper layer and local host device hardware like NIC (Network Interface Card).
- NIC: It is the physical hardware component which allows a device to interact with wired/wireless network.

ii) Media Access Control (MAC):

- It is responsible for data encapsulation process and controlling access to the network devices.
- It is also responsible for inserting physical address information (MAC address) on the header of the frame.
- MAC address is also called as Burned in Address (BIA).
- The MAC address size is 48 bits.
- The MAC address on NIC is considered to be unchangeable as the vendor of device had provided during the fabrication. Hence it is called as BIA address.

Characteristics: (i) Size = 48 bits

(ii) Not Case-Sensitive

(iii) Written in Hexadecimal

- The first 24-bits of MAC address is called as Organizationally Unique Identifier.
- The next 24-bits are uniquely addressed, i.e. Device Specific
- MAC layer handles error detection by inserting trailer in the frame.

⇒ Frame Structure of the Data-Link Layer (Ethernet header):

Preamble (7 bytes) & Start Frame Delimiter [SFD] (1 Byte)	Destination MAC address	Source MAC address	Type/Length	Data	FCS
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8 bytes

6 bytes

6 bytes

2 bytes

16-1500 bytes

4 bytes

(48 bits)

FCS: Frame Check Sequence

→ Type/Length: It provides information about IP version-4 & IP version-6.

→ Data: All ethernet frame are required to be atleast 64 bytes in length.

→ FCS: Used for error detection, maintains the integrity of the frame.

Minimum Length = 64 bytes (6+6+2+46+4)

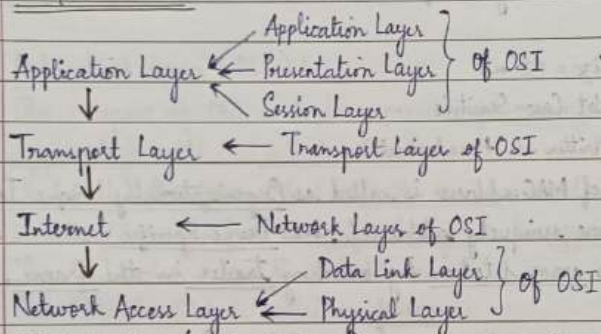
Maximum Length = 1518 bytes (6+6+2+1500+4)

• Frames that are less than 64 bytes are called as RUNTS. In order to make it 64 bytes, additional bits known as PAD are inserted to increase the size of the frames to the minimum length.

• Frames that are more than 1518 bytes are called as JUMBO/GIANT FRAME.

Physical Layer (PL):

- It is the electrical wire, media type, connections such as ports interface on a network.
- It ensures following elements so that data can be transmitted over a network:
 - Physical Components: Cables, NIC → all hardware components for communication.
 - Encoding Process: It defines the method used by a device to convert stream of messages into code.
 - Signaling Element: Transferring signals wireless.

TCP/IP Model:4/5 layers
(Both correct)11-09-24 # IPv4 Header: (Network Layer)Version:

- Size is 4 bits.
- It identifies the message as IPv4 message.
- Internet Header Length: (IHL)
- The size is 4 bits.
- This bit indicates where the header section ends & data section starts.

Differentiated Service (DS):

- The size is 1 Byte / 8 bits.
- It determines the priority of the packet.
- The first 6 bits is called as differentiated service code point (DSCP).
- The last 2 bits are called as Explicit Congestion Notification (ECN).

Total Length: (TL)

- The size is 16 bits.
- It tells us about the length of the IPv4 packet.

Identification: (ID)

- The size is 16 bits.
- Used to provide identification numbering of each fragmented packet that belongs to original packet.

Flag: (FLAGS)

- The size is 3 bits.
- It gives us information whether the packet is to be fragmented or not.

Fragmented Offset: (FRAG OFFSET)

- The size is 13 bits.
- It is used to indicate the sequencing position of a fragmented packet.

Time-to-Live (TTL):

- The size is 8 bits.
- It is used to determine the life of the packet in the network.

Protocol: (PRO)

- The size is 8 bits.
- It is used to indicate the payload type that is present in the packet.

Header Checksum: (CHKSUM)

- The size is 16 bits.
- It indicates whether there is any corruption within the header.

Source IP Address: (SRC IP)

- The size is 32 bits.

Destination IP Address: (DST IP)

- The size is 32 bits.

IPv6 Header: (Network Layer)

Version:

- The size is 4 bits.

Traffic Class:

- The size is 8 bits.
- Functions are same as differentiated services (DS) in IPv4.

Flow Control: (Flow Label)

- The size is 20 bits.
- It is used to inform the router on the network to use the same type of handling.

Payload Length:

- The size is 16 bits.
- It represents the length of data in IPv6 packet.

Next Header:

- The size is 8 bits.
- It indicates the payload type that is present in the packet.

Hop Limit:

- The size is 8 bits.
- Same as Time-to-live (TTL) in IPv4.

Source IP Address:

- The size is 128 bits.

Destination IP Address:

- The size is 128 bits.

TCP Header: (Transport Layer)

Source Port:

- The size is 16 bits.

Destination Port Number:

- The size is 16 bits.

Sequence Number:

- The size is 32 bits.
- It is used to re-assemble the process on the receiver's side.

Acknowledgement Number:

- The size is 32 bits.

Header Length: (Data Offset)

- The size is 4 bits.
- It indicates the length of TCP.

Reserved:

- The size is 6 bits.
- It is used for future application.

Control Bits:

- The size is 6 bits.
- It is used to specify various TCP flags such as **URG** (Urgency of TCP segment), **ACK** (Acknowledgement), **PSH** (Used to reset a connection), **SYN** (Synchronization), **FIN** (Finish/Terminating the connection).

Window:

- The size is 16 bits.
- It is used to indicate the number of bits that can be accepted during the transmission between sender & the receiver.

Checksum:

- The size is 16 bits.
- It is used for detecting error.

Urgent:

- The size is 16 bits.
- It is used to indicate urgency on the TCP Header.

Option:

- Optional in nature.

Application Layer Data:

⇒ Topology: It is the layout of a network.

⇒ Uses of Topology:

- i) It helps to understand how system and devices are interconnected. It also provide information about the area of improvement.
- ii) It also helps to identify the fault present inside the network.

Topology $\leftarrow$ Physical Topology
Logical Topology.

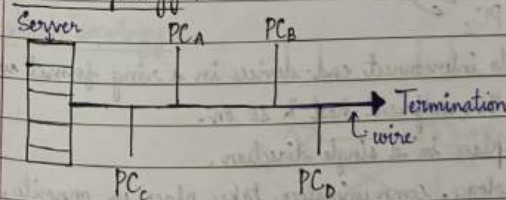
⇒ Physical Topology:

- It contains high level overview, sharing the actual arrangement and placement of devices within an organization such as geographic location.
- It also shows the type of network connection and connectors and how they are connected between devices.

⇒ Logical Topology:

- Low level diagram that focus on technical details of the network such as IP address.

1. BUS Topology:



$$\text{Number of links} = n + 1$$

n = no. of PC's

1 = Backbone

- No longer used in modern networking.
- Single networking cable is used as the backbone of entire network.
- All nodes are connected to the same networking cable which allows each node to access the resources on the server or those that are shared from one node to another.
- At the end of the backbone networking cable, terminators are installed to terminate/ground the cable if there are any unwanted electrical signals on the wire.

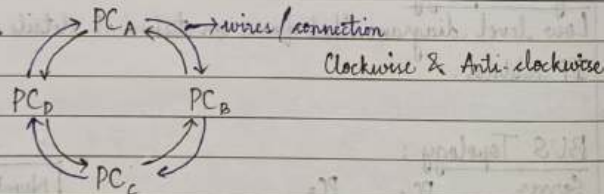
- In case of shared ethernet, if a message is sent to a specific destination MAC address, each node on the physical segment receives the message but only the MAC address embedded in the NIC matches the destination MAC will accept the message.

Advantages:

- Very simple.
- Cost effective (Single Transmission line).

Disadvantages:

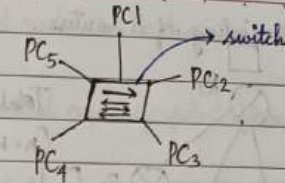
- Lots of overhead and excessive traffic on the network.
- If more than one device attempt to access the network and send message, this creates collision.
- Since it contains a single backbone cable, if there is break at any point in the cable then the entire network will stop working.

2. Ring Topology:

- This topology is designed to interconnect end-devices in a ring format where one host is connected to another adjacent host & so on.
- Here, communication takes place in a single direction.
- In case of dual ring topology, communication takes place in opposite direction on each ring i.e. Clockwise & Anti-Clockwise.

Advantage: i) It is simple.Disadvantages:

- Communication takes place in a single direction, i.e. message have to be forwarded till it reaches to its intended receiver.
- At a time, only one host can send a message.
- Since all nodes are interconnected, if a node within the ring topology is not available, then the entire network goes down.

3. Star Topology:

Hub = 06 ports
Switch = 24 ports

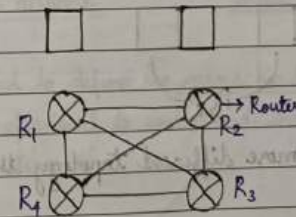
- It allows the host to connect to a single networking device like switch.
- This topology ensure that if any host on the network want to send a message to another host, the sender will forward the message to the network switch, which forward the message to the intended destination host.
- If one host on the network is offline/unavailable, the entire network is not affected.
- If switch is down or unavailable, then the entire network or all the connected host to the switch will be affected and thus communication will not take place.

Advantages:

- It is easy to support scalability for organization that are increasing the number of devices that can be connected to the network.

Disadvantage:

- If the switch is not working then the entire network is down.

4. Mesh Topology:

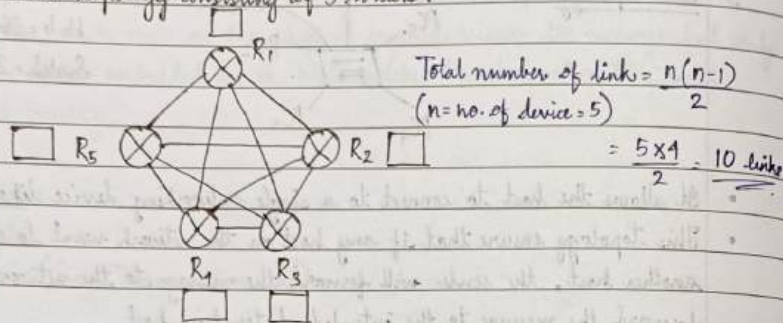
Total number of link = $\frac{n(n-1)}{2}$
(n = no. of devices)

$$= \frac{4(3)}{2} = 6$$

(all)

- Each router is connected with all other routers.

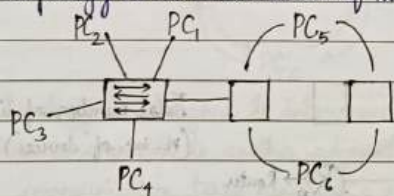
Q) Design a mesh topology consisting of 5 routers.



- Each device within the network is connected to all other device, creating a physical and logical mesh design.
- This topology provides redundancy, because of which the complexity and the expenditure increase.
- While having multiple redundant connection from one node to all other node in a mesh topology, this type of topology can be complex to troubleshoot if an issue occur due to the number of available path.

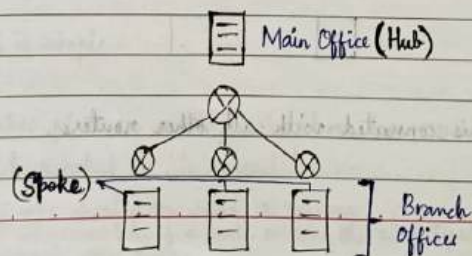
5. Hybrid Topology:

- This topology is a combination of two or more topology.



- Combination of two or more different topology that are interconnected to create a network.

6. Hub & Spoke:

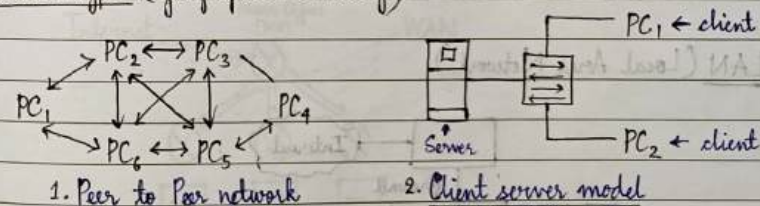


- It provides a centralised node that act as the main hub for all other nodes to interconnect.
- The hub acts as the central referring point for all traffic between other nodes on the network.
- Each node is connected to the hub and these connections are referred as SPOKE.
- This topology doesn't allow nodes to directly connect, rather all network traffic is sent through the hub, within this topology which is responsible for forwarding traffic to the other node on the network.

⇒ Difference between Hub and Spoke & Star Topology:

- Hub and Spoke are more scalable as compared to that of star topology.
- Hub and Spoke are more fault tolerant, as a hub failure will only affect the connected devices, while a central device failure in star topology affect the entire network.
- Hub and Spoke is more complex, scalable and fault tolerant version of star topology.

Network Types: (geographic boundary)



- Network type are used to define the geographic boundary or limitation of a network and help network professional to understand the relationship between the connected devices within the network.

1. Peer to Peer Network:

- Each client is logically connected and share their resources with other client on their network without the need of centralised, dedicated server.
- Since there is no centralised management of any network resources or services, peer to peer networking doesn't support scalability well.

E.g. Sharing file using a Torrent.

Seeds | Seeders
Leeches | Leechers

- Seeds (or) seeders are the device that have the complete file on their device and share it with others to download.
- Leeches (or) leechers are the device that download the file from the seeds.
- Peer to Peer network - Security is also concerned.

2. Client-Server model :

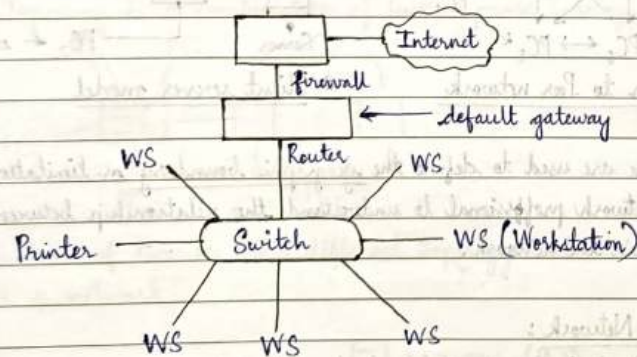
- It provides resources to the devices present in a network.
- Clients are the devices that are requesting service from the server.
- Servers are the devices that respond to the request of client.

Advantages : i) Scalability
ii) Security

3. PAN (Personal Area Network)

- It is a small network created by a user to interconnect device like laptop, smart phone, printer, etc.
- A user can setup a wired or wireless network to ensure their devices are connected to the same network.

4. LAN (Local Area Network)

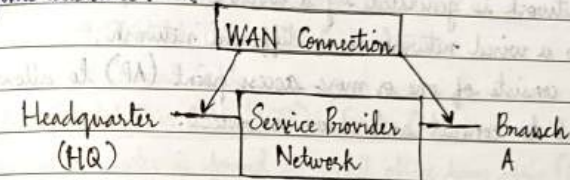


- The switch that is connected to the router which functions as the default gateway to forward the traffic from the internal network to a remote network i.e Internet.
- Without a default gateway, the client won't be able to communicate with remote network.

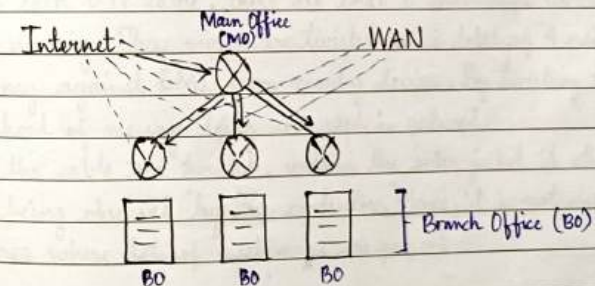
- The router connects the firewall to filter inbound and outbound traffic between the internal network and external network (Internet).

5. WAN (Wide Area Network)

- It allows organization to extend their LAN and its resources over a large geographic distance.



- WAN connection is established by a managed service provider / Internet Service Provider. They are responsible for establishing and maintaining the WAN network for the organization.
- All traffic sent into a WAN connection by a organization remains private as the traffic passes through the Service Provider's Network Infrastructure.



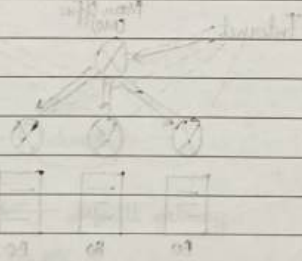
- WAN solution will be used to share the network resources between the branches, while Internet service provide access to the online services and resources that are outside the company's network.
- The above network is not cost efficient as the organization is charged for the Internet service at each Branch location as well as Main Office.
- The better solution is to implement a single Internet connection is only provided to the Main Office.
- The Internet service can be re-distributed within the WAN connected to each Branch router.

6. MAN (Metropolitan Area Network)

- Cost is less and high speed connection to the internet.
- Limited to a city/organization.

7. Wireless LAN

- This type of network is generated by a wireless router, while the wireless router is connected to a wired network or a different network.
- Wireless Router consists of one or more access point (AP) to allow wireless capable client to connect and share resources.



Types of Connections (wired):

- Copper cable
- Fiber Optic cable

i) Copper cable: → Unshielded Twisted Pair Cable (UTP)
→ Shielded Twisted Pair Cable (STP)
→ Coaxial Cable

→ Unshielded Twisted Pair Cable (UTP): (Copper Cable)

- This type of connector is found in small office home office (SOHO) network.
- They are referred as unshielded because the physical cable doesn't contain a protective coating to prevent the actual conductor from observing electromagnetic interference (EMI) from machineries and other components.
- As the electrical signals that is obtained from the Network Access Layer are transmitted along a copper wire, can observe EMI from nearby devices which can corrupt the signal/electrical signals.
- Within each UTP cable, there are total 8 individual wires with a unique colour coating. These wires are twisted in a total of 4 pair to improve the resiliency against EMI from nearby devices. By twisting the conductor, the likelihood of message being corrupted is reduced.
- While the cable are twisted, within the outer jacket it still doesn't stop EMI from being observed by the conductor. Hence, it is not used/recommended in the area where lots of machineries are present.

→ Shielded Twisted Pair Cable (STP): (Copper Cable)

- The STP cable has shielding i.e. foil shielding around the twisted pair of wires. The protective shielding is used to prevent EMI from entering the copper conductor within the cable.
- STP cable is more expensive than UTP because of additional layer of protection.
- The outer jacket of both UTP and STP are made up of PVC (Poly Vinyl Chloride), a material that releases toxic/harmful fumes to human when burnt. That's why such type of cables are called as Non-Plenum rated cable.

Layers of protection for Plenum-rated cable:

- It is recommended to implement Plenum-rated cable within the plenum space of the organization/building.
- The plenum rated cable contain a Plenum grade outer jacket to prevent fire burning the conductor.
- A pair of separate spines are present in order to keep each pair of twisted cable separate from other pair.
- This type of cable are less toxic when they are burnt during a fire.

Attenuation:

While a network cable may specify a maximum distance in reality, these copper cable experience attenuation (loss of signal as it travels along a medium from source to destination) as the electrical signal travel along the wires.

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→ Coaxial Cable: (Copper cable)

Basic coverings of coaxial cable:

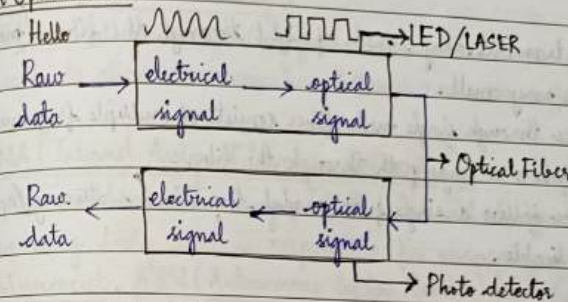
- It is made up of copper.
- It has a foil coating around the copper core to protect it from EMI (Electromagnetic Interference).
- This cable contain multiple layers of protection to prevent harm to the copper conductor at the core.
- Various layers of coaxial cable (outer to inner layer):

- Outermost layer
- Protective Coating (PVC):** It protects the internal media from external elements and environmental factors.
 - Braided Shielding:** It prevents EMI from reaching the copper conductor.
 - Foil Shielding:** It prevents radio frequency interference (RFI) from reaching the copper conductor.
 - Dielectric Insulator:** This insulator separates the copper conductor and the Braided shielding, and also to keep the copper conductor at the centre of the coaxial cable.

Innermost layer

- Copper Conductor:** It contains electric signals.

ii) Fiber Optics Cable:



Advantages of Fiber Optics Cable over Copper Cable:

- Fiber Optics Cable are immune to EMI.
- Attenuation over Fiber Optics Cable is lower than that of Copper Cable.
- Fiber Optics Cable allows faster throughput of data over the cable since fiber optics cable uses LED/LASER to transmit data in the form of light rather than electrical signals. (Photons travel at a higher speed as compared to that of electrons.)
- Throughput:** It is the rate at which data are processed or transmitted over the network.

Disadvantages of Fiber Optics Cable over Copper Cable:

- Fiber Optics Cable is expensive as compared to that of Copper Cable.
 - Fiber Optics Cable is very fragile and easy to break as the core is made up of GLASS (or) PLASTIC.
- Fiber Optics Cable transmit data using pulses of light that are sent down a thin CORE of plastic (or) glass, which is surrounded by a material called as CLADDING.
 - The combination of core & cladding allows for the light to be transmitted through the process of Total Internal Reflection (TIR).

Types of Optical Fiber:

- Single Mode Fiber (SMF)
- Multi Mode Fiber (MMF)

a) Single Mode Fiber:

- It is constructed to transmit only 1 mode of light through the fiber. (mode $\rightarrow$ path)
- The core diameter is very small.
- The light that passes through single mode fiber consists of multiple frequency, but all of these frequencies follow a single path through the fiber.
- Since all light waves follow a single path, modal dispersion (scattering/spreading of light) are not applicable.
- Attenuation is low.
- Optical Source used is LASER.
- More expensive than Multi Mode Fiber (MMF).
- It operates at a wavelength of $\lambda = 1310 \text{ nm}$ to $\lambda = 1550 \text{ nm}$

b) Multi Mode Fiber:

- Larger diameter than SMF.
- Modal dispersion takes place.
- Cheaper than SMF.
- Instead of LASER, LED is used.

$\Rightarrow$ The need for IP addressing:

- An IP address is a layer 3 logical address that is assigned to all devices on a network to allow communication between nodes in different IP network.

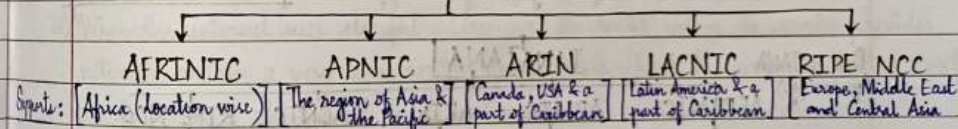
IANA (Internet Assigned Numbers Authority):

- In the world of networking, IANA (Internet Assigned Numbers Authority) is the governing body that is responsible for managing global IPv4 & IPv6 address assignment, ASN (Autonomous System Number) allocation to organization that manages a large number of public network, Protocol Assignment and Root Domain Name System directories and services of the world. (DNS)

- To assist with distributing IP address to ISP around the world, IANA delegates the responsibility of IP distribution to 5 RIR (Regional Internet Registries) around the world.

- Each RIR is responsible to distribute the IP address to specific geographical location.

IANA (RIR's).



AFRINIC - African Network Information Center.

APNIC - Asian Pacific Network Information Center.

ARIN - American Registry for Internet Number.

LACNIC - Latin America & Caribbean Information Center.

RIPE NCC - Research IP European Network Coordination Center.

- The IANA assigns ASN (Autonomous System Number) to the various RIR around the world. Then the RIR allocates these AS Numbers to the network operators such as ISP & ASN.

- The ISP uses their AS Number to share their network routing prefix with other

- ISP using the Border Gateway Protocol (BGP).

- BGP is an exterior gateway protocol that allows network operators such as ISP to exchange their routing information with each other, allowing user on the internet to reach network beyond their local ISP.

- Each ISP is being provided with an unique AS Number. Internally network operators use various Interior Gateway Protocol (IGP) to forward the packet within a network.

⇒ Public vs Private address space:

- IANA creates two separate IPv4 address space:

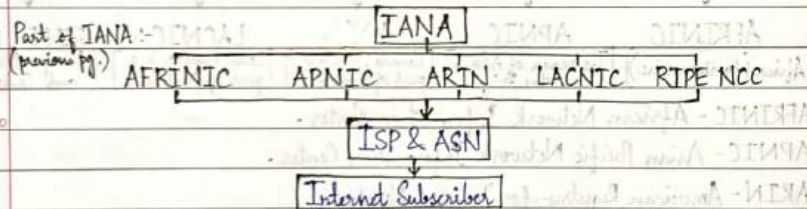
- Public address space
- Private address space

i) Public address space:

- It provides IPv4 addresses that are routeable on the internet and can be assigned to the devices that are directly connected to the internet.

ii) Private address space:

- Specifies the classes and range of IPv4 address that are non-routeable on the internet and can be assigned to the devices on private network only.
 unable to communicate with the outside world



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Class	Range	Default Subnet Mask	No. of N/W	No. of usable IPv4 address
A	0.0.0.1 - 127.255.255.255	255.0.0.0	126 (2^7-2)	16,777,214 ($2^{24}-2$)
B	128.0.0.1 - 191.255.255.255	255.255.0.0	16384 (2^{14})	65534 ($2^{16}-2$)
C	192.0.0.1 - 223.255.255.255	255.255.255.0	2097152 (2^{21})	254 (2^8-2)
D	224.0.0.1 - 239.255.255.255	NA		
E	240.0.0.1 - 255.255.255.255	NA		

e.g. of Private Address space (not actual range)

A	10.0.0.1 - 10.255.255.255
B	172.16.0.1 - 172.31.255.255
C	192.168.0.1 - 192.168.255.255

IPv4
 ↳ Classful Address
 ↳ Classless Address

Classful Address:

- It describes how each IPv4 address class contain a specific number of networks. Here each network contains a specific number of usable IPv4 address to host devices.

Classless Address:

- Classless Address removed the need to use of default subnet mask for a specific range of IPv4 address.

⇒ Disadvantages of Classful Address:

- Wastage of unused IP addresses within an organization.
- It was not flexible to support organization of various sizes.
- When the internet uses classful addressing, the router within the service provider network contain a very large routing table, which affect the performance of a router on forwarding the packet to its destination.
- When working with classful address, everyone needed to use the default subnet mask with corresponding address block that was assigned to their organization. Companies were not able to locally segment their network into multiple subnetwork from a single network block, which was not flexible for business of different size.

* In order to reduce the wastage of public IPv4 address, IANA created the private IPv4 address space, which allows organization to use a specific set of IPv4 addresses on private network.

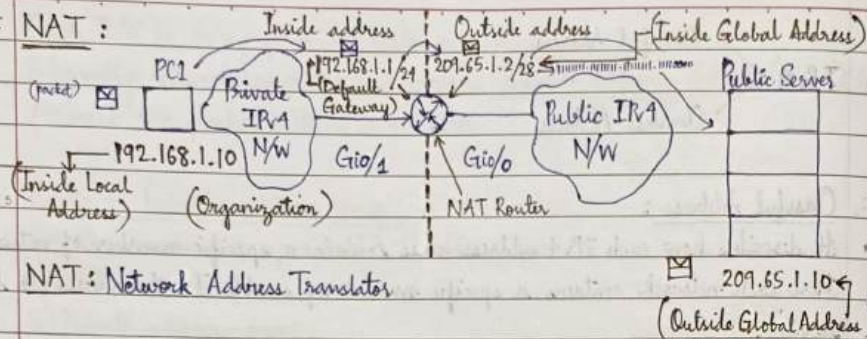
* These addresses are non-routeable on the internet i.e. the device assigned with IPv4 address will not be able to communicate directly with devices on the internet and vice-versa.

* ISP implement security mechanism such as: Access Control List (ACL) to filter traffic originating from their customers network that have source private address, therefore separating the private and public address space from meeting each other.

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Date 03/10/24

NAT :



⇒ The steps for NAT operation:

1. PC1 creates a packet with source IPv4 address of 192.168.1.10 (Inside local Address) and a destination address of 209.65.1.10 (Outside Global Address) and forward the message to the default gateway i.e. 192.168.1.1.
2. The router inspect the source and destination IP addresses within the IP header of the packet to determine the destination and check its routing table for a route/path.
3. Before the router forward the packet to the internet, it translates the source private IPv4 address from 192.168.1.10 to the public IPv4 address on the router as 209.65.1.2 (Inside Global Address).
4. The public server will see the source IPv4 address of the packet as 209.65.1.2 (Router) but not the address of PC1 because NAT allows the private network to be hidden behind the public IPv4 address of the Router.

⇒ Advantages of using NAT on network:

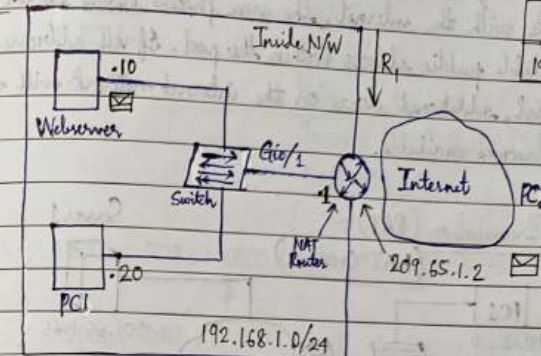
- i) It helps to conserve public address (IPv4) space by allowing organization to use private address on their internal network.
- ii) Translates private address to public address.
- iii) The entire private network can be hidden by a single public address.
- iv) Network professionals can maintain consistency in their private addressing schemes within their companies.

⇒ Disadvantages of using NAT on network:

- i) Since translation takes place, network performance is affected (delay takes place).
- ii) NAT modify addresses within the packet, so end to end connectivity is lost between the sender & the receiver. This makes it difficult for the receiver to determine the true source address.

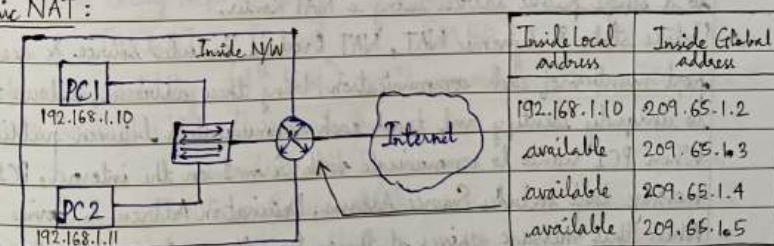
⇒ Types of NAT:

i) Static NAT:



- Static NAT allows network professionals to create a one-to-one mapping between an inside local address such as a private IPv4 address on an internal server to an inside global address such as the public IPv4 address on the router.
- Network professionals use static NAT to create a one-to-one mapping between 192.168.1.10 (Private IP Address) and 209.65.1.2 (Public IPv4 Address) on the router.
- The user on the internet i.e. PC2 will not see the Private IPv4 Address on the internal network because it is being hidden behind the public IPv4 address on the router.

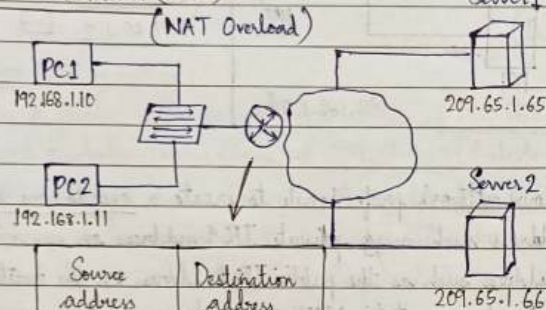
ii) Dynamic NAT:



- Dynamic NAT allows network professionals to create many-to-many mapping between multiple private IPv4 addresses & public IPv4 addresses.
- In dynamic NAT, a pool of public IPv4 addresses are assigned to the private IPv4 addresses on a first-come-first served basis.
- When PC1 wants to communicate with a device on the internet, the router (NAT) will check the NAT pool for an available public IPv4 address to translate the source private address into the NAT pool to the destination on the internet.

While PC1 is using the 209.65.1.2 address, if another device such as PC2 wants to communicate with the internet, the same process occurs on the router and use the next available public address within the pool. If all addresses within the NAT pool are being used, additional device on the internal network will need to wait till an address becomes available.

iii) Port Address Translation (PAT):



Source address	Destination address
192.168.1.10.1234	209.65.1.65.80
192.168.1.11.1235	209.65.1.66.80

- PAT allows organization and home user to perform many-to-one translation. It allows multiple devices with private addresses to translate their source address to a single public address using a NAT Router.
- Unlike static & dynamic NAT, NAT Overload includes source & destination service port number of each communication. Using these addresses, allows the NAT Router to uniquely identify and track each communication between public & private N/W.
- When PC1 wants to communicate with Server1 on the internet, PC1 creates a message that includes Source Address, Destination Address & Service Port Number.
- When this message arrives at the router, the router records the Source Address and the Service Port Number (SPN). Then the router translates the Source Private Address to Source Public Address keeping the Source Port Number unchanged.

⇒ Structure of IPv4 & IPv6:

IPv4
32 bits

IPv6

128 bits

Written in hexadecimal (0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F)

Grouped into 8 hexets (16 bits)

8x16 = 128 bits

Eg. for IPv6: 2001:0DB8:0000:1111:0000:0000:0000:0200

2001:DB8:0:1111:0:0:0:200 → 2001:DB8:0:1111::200

0010000000000001 0000110110110000

When there are 2 or more hexets that are all 0, a double colon is used to replace the consecutive 0 within the IPv6 address.

- As with all IPv4 addresses, there is a subnet mask to accompany it. The same is also applied to IPv6 addresses, the default subnet mask (or) prefix length of IPv6 address is /64.

⇒ Address Structure of IPv6:

2000:0DB8:0000:1111	0000:0000:0000:0200
Global Routing Prefix	Subnet Interface ID
[48 bits]	[16 bits] [64 bits]

- Information provided by (4th hexet) - It represents the Interface ID portion Service Provider like ISP. - It represents the subnet ID used by ISP to create subnetwork of the network block.

- Thus, combining the Global Routing Prefix, Subnet ID and Interface ID, a client is being assigned with unique 128 bits IPv6 address on its Network Interface Card (NIC).

⇒ Types of IPv4 & IPv6 address:

i) APIPA (Automatic Private IP Addressing):

- When a client is connected to a network, the device is automatically assigned an IPv4 (or) IPv6 address from a Dynamic Host Configuration Protocol Server (DHCP Server) [If there is a large organization] or modem/router provided by the ISP which provides the DHCP Server for the client (In case of small network).
- If the client doesn't receive an IP address from the network because of (client is unable to communicate with the DHCP Server or DHCP Server is offline or client is configured to use a static IP address), in such situation, the client is automatically assigned itself a special unique IP address ranging from 169.254.0.1 to 169.254.255.254 with default subnet mask of 255.255.0.0.
- This feature is known as APIPA also known as Link-Local address. Such types of feature are adapted in Linux, Microsoft, etc.

ii) EUI-64 (Extended Unique Identifier)

- DHCP version 6 Server is used to provide address of IPv6 to the client.
- DHCPv6 Server keep record of each client that receives an IPv6 address and the Lease time, that's why it is called as Stateful address.
- In some IPv6 network a Stateless IPv6 technology is known as Stateless Address Auto-Configuration (SLAAC) that helps the client to obtain Global Unicast IPv6 Address on the network.

⇒ Difference between SLAAC & DHCPv6:

i) SLAAC doesn't keep a record of each client's IPv6 address whereas DHCPv6 keeps it.

ii) When using SLAAC on an IPv6 network, SLAAC provides the network portion of IPv6 address to the client (64 bits) i.e. first 64-bits of the IPv6 address. The client use a process Extended Unique Identifier-64 to convert its 48-bits MAC address to 64-bit address.

E.g. for SLAAC: 2001:DB8:0:1111::/64

Network portion of IPv6 address

Example for SLAAC:

2001:DB8:0:1111::/64

Network portion of IPv6 address

i) 48 bit MAC address - FC 99 47 75 CE E0

11111100	10011001	01000111	01111011	11001110	11100000
----------	----------	----------	----------	----------	----------

Organization Unique Identifier (OUI) Device

48-bit MAC address → 64 bits ID of IPv6.

ii) Insert hexadecimal value of FF FE in the middle of MAC address.

FC 99 47 FF FE 75 CE E0

11111100	10011001	01000111	11111111	11111110	01110101	11001110	11100000
----------	----------	----------	----------	----------	----------	----------	----------

FF FE: This value is reserved by IEEE & can only appear in a EUI-64 generated from a 48-bit MAC address.

iii) Flip the 7th bit within the first byte so that 0 becomes 1 (or) 1 becomes 0. This bit indicate if the NIC is administered locally (0) or is globally unique (1).

11111110 10011001

iv) Convert the binary into hexadecimal to view the EUI-64 portion of the address.

FE 99 47 FF FE 75 CE E0

v) Putting it altogether, the EUI-64 bits IPv6 address that will be assigned to the device network is

⇒ 2001:DB8:0:1111:FE99:47FF:FE75:CEED

⇒ 2001:0DB8:0000:1111:FE99:47FF:FE75:CEED (128 bits)
(Middle section of Interface ID)

Q) Which of the following is a EUI-64 address?

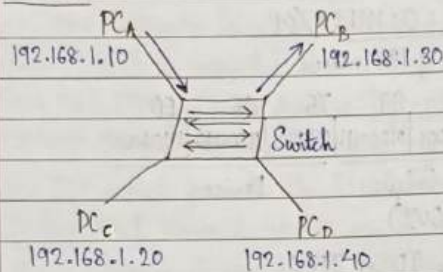
(i) 2001:DB8:0:1111:FE99:47FA:FE75:CEED

(ii) 2001:DB8:0:1111:FE99:47FF:FE75:CEED

(iii) 2001:DB8:0:1111:FE99:47FB:FE75:CEED

(iv) 2001:DB8:0:1111:FE99:47FC:FE75:CEED

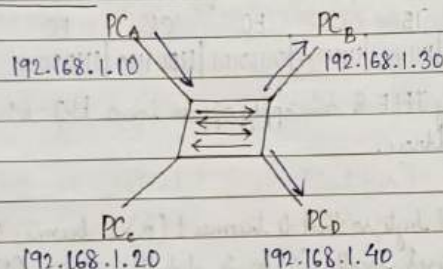
iii) Unicast:



→ one-to-one

→ provided by both IPv4 & IPv6.

iv) Multicast:



→ one-to-many

→ provided by both IPv4 & IPv6.

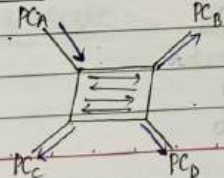
- PC_A uses multicast address to send message to all the device that belongs to the same multicast address group.

- Multicast IPv6 address have a network prefix of FF00::/8.

Different types of Multicast address:

- FF02::1/12 - indicates all IPv6 enabled node on a network.
- FF02::2/12 - indicates all IPv6 enabled router on a network.
- FF02::5/12 - indicates all router that are running Open Shorted Path First (OSPF) dynamic routing protocol.
- FF02::A/12 - indicates all router that are running the Enhanced Interior Gateway Routing Protocol (EIGRP) dynamic routing protocol.

v) Broadcast:



→ one-to-all

→ present only in IPv4 not IPv6.

* Subnet Mask - 255.255.255.0

* IP Address - 192.168.1.255 (known as Broadcast Address of IPv4)

vi) Anycast:

→ one-to-nearest.

→ present only in IPv6.

• When a client sends a message to an anycast IPv6 address, the message is delivered to the geographically nearest server which is configured with the anycast address.

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Relating into IPv6 concepts:

i) Tunneling:

- Tunneling allows IPv6 packet to be transported over an IPv4 network. The forwarding router is responsible for encapsulating the IPv6 packet inside an IPv4 packet before sending it over the IPv4 network. This type of tunneling is referred as 6-4 tunneling.
- In case of 4-6 tunneling IPv4 packet will be encapsulated in IPv6 packet so that it can be forwarded to IPv6 network.

ii) Dual Stack:

- Dual stacking both IPv4 & IPv6 address on the same NIC of an end device such as router which will allow the device to communicate efficiently on both IPv4 & IPv6 network.

iii) Translation:

- NAT 64 allows devices within an IPv6 network to communicate with host of an IPv4 network using an address translator whose operation is similar to NAT for IPv4 network devices.

iv) Router Advertisement:

- Device on an IPv6 network automatically obtain a Global Unicast Address (GUA) using ICMPv6 messages.
- Whenever an IPv6 enabled device is connected to IPv6 network, it sends a Router Solicitation (RS) message to the network to discover any IPv6 router.
- An IPv6 router (enabled router) respond with a Router Advertisement (RA) message which is used to inform the host on the network - how to obtain a GUA IPv6 address.

• The RA message provides the following network information to the client:

- The network prefix & the prefix length of the address.
- The default gateway of IPv6 address.
- The DNS server IP address & the Domain Name.
- SLAAC methods for configuring GUA IPv6 address for the client.
- SLAAC with Stateless DHCPv6.
- Stateful DHCPv6.

v) SLAAC

→ ④ • SLAAC allows devices to be configured on a network with GUA IPv6 address without the need of DHCPv6 server.

• The steps are:

- Client send RS message to seek/ find an IPv6 router.
- The router respond with RA message & provide the network prefix & prefix length.
- The client use EUI 64 process to convert 48 bit MAC to 64 bit Interface ID. The 64 bit Interface ID is appended with 64 bit network ID to make it 128 bits IPv6 address.

→ ⑤ • Additionally the RA message from the router can indicate to the client to use both SLAAC & Stateless DHCPv6 to obtain a GUA address. In this situation, the RA message inform the client the following information/instruction:

- Use SLAAC to create its own GUA address.
- Use the router's local address as the default gateway of the network.
- Use the stateless DHCPv6 server to provide DNS Server Address & the Domain Name.

→ ⑥ • Another method for the client to receive the address is by using Stateful DHCPv6 server. It provides GUA IPv6 address, Prefix length, DNS Server Address and Domain Name.

- The router provides the default gateway address as its Link Local Address.

Important characteristics of subnet mask:

- The length of subnet mask for IPv4 is 32 bits & for IPv6 is 128 bits.
- It helps devices to identify the network & host portion of the IP address.
- It helps devices/network professionals to determine the total no. of IP addresses and usable address within an IP network.
- It helps a sender device to determine whether the destination host is on the same network or on another network.

Default Subnet Mask Example

A	255.0.0.0/8	10.5.0.1/8
B	255.255.0.0/16	128.168.11/16
C	255.255.255.0/24	192.168.11/24

↳ x → N/w Prefix

→ /24 or /64 or /x are referred as Network Prefix & represents the subnet mask in simplified form.

→ x value is calculated based on total no. of bits which are '1' within the Subnet Mask of IPv4 or IPv6 address.

Class B: 172.16.0.0

=	[10101100.00010000]	[00000000.00000000]
Subnet Mask:	[11111111.11111111]	[00000000.00000000]
	Network Portion	Host Portion

• The Network Portion of the IP address is the same for all devices within the same network, while the Host Portion of the IP address is unique to the interface of the end device only.

• The '1's' within the Subnet Mask that are aligned with the bits within the IP address are used to identify the Network Portion of the device, while the '0's' within the Subnet Mask that are aligned to the remaining bits within the IP address indicate the Host Portion of the address.

Subnet Mask: ① 255.255.224.0/x

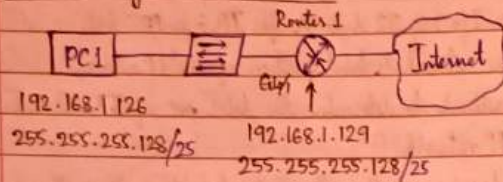
11111111.11111111.11100000.00000000

No. of 1's = 8+8+3 = 19 ⇒ x=19

② 255.255.255.240/x

11111111.11111111.11111111.11110000

No. of 1's = 8+8+8+4 = 28 ⇒ x=28

Determining the Network ID: ^{identify a particular network}

PC:

IP: 192.168.1.126

= 11000000.10101000.00000001.01111110

(AND operation)

SM = 11111111.11111111.11111111.10000000 (255.255.255.128)

NW ID = 11000000.10101000.00000001.00000000 = 192.168.1.0 / 25

Router 1:

IP Address: 192.168.1.129

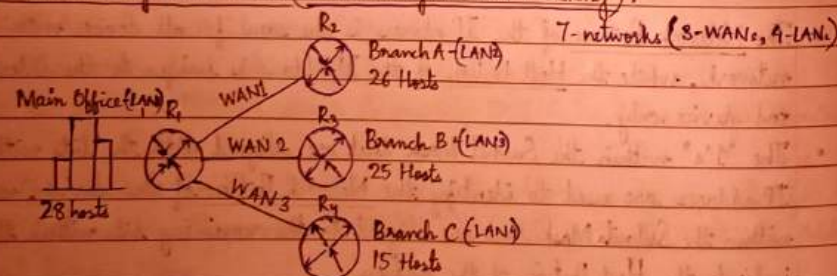
= 11000000.10101000.00000001.10000001

Subnet Mask: 11111111.11111111.11111111.10000000 (255.255.255.128/25)

Network ID: 11000000.10101000.00000001.00000000 = 192.168.1.128/25

Conclusion: They both have different Network ID (PC & Router 1), so communication will not take place.

06-11-24 # IPv4 subnetting and VLSM (Variable Length Subnet Masking):



- Using a classless addressing scheme, allows network professionals to create smaller network with custom subnet mask with fewer usable IP addresses to avoid wastage by using the technique called as Subnetting.

⇒ Benefits of subnetting:

- Efficiently distributing IP addresses with least wastage.
- To create more network with smaller broadcast domain.

Subnetting:S-1: Determining the appropriate IPv4 block: → Class C

- Determining the no. of networks and the size of the largest network within an organization help us to choose an appropriate address class for the organization.
- Class A & Class B result in wastage of addresses. Thus, using Class C network block may seem to be the most appropriate as it's the small address class with 254 usable IPv4 addresses.
- When creating subnet, the size of each subnet should be able to fit the largest network within an organization.

S-2: Creating new subnet (subnetwork):

- In order to create new subnets from an address block, we convert some of the host bits into new network bits. This allows us to create more network while reducing the number of IPv4 addresses that are available within each subnet.

Subnet 1: 11000000.10101000.00000000.00000000 → 192.168.0.0/27

Subnet 2: 11000000.10101000.00000000.00100000 → 192.168.0.32/27

Subnet 3: 11000000.10101000.00000000.01000000 → 192.168.0.64/27

Subnet 4: 11000000.10101000.00000000.01100000 → 192.168.0.96/27

Subnet 5: 11000000.10101000.00000000.10000000 → 192.168.0.128/27

Subnet 6: 11000000.10101000.00000000.10100000 → 192.168.0.160/27

Subnet 7: 11000000.10101000.00000000.11000000 → 192.168.0.192/27

Subnet 8: 11000000.10101000.00000000.11100000 → 192.168.0.224/27

S-3: Assigning subnets to each network:

Consider subnet 1 to be the main office.

(Network ID) Subnet 1: 11000000.10101000.00000000.00000000 → 192.168.0.0/27

(WAN ID+1) 1st usable IP: 11000000.10101000.00000000.00000001 → 192.168.0.1/27

(Broadcast - 1) last usable IP: 11000000.10101000.00000000.00011110 → 192.168.0.30/27

(Subnet 2 - 1) Broadcast: 11000000.10101000.00000000.00011111 → 192.168.0.31/27

Consider Subnet 2 to be the main office:

Subnet 2 : 11000000.10101000.00000000.00100000 → 192.168.0.32/27

1st usable IP: 11000000.10101000.00000000.00100001 → 192.168.0.33/27

Last usable IP: 11000000.10101000.00000000.00111110 → 192.168.0.62/27

Subnet 3-1) Broadcast: 11000000.10101000.00000000.00111111 → 192.168.0.63/27 (Subnet 3-1)

(Next N/W ID - 1)

Consider Subnet 3 to be the main office:

Subnet 3 : 11000000.10101000.00000000.01000000 → 192.168.0.64/27

1st usable IP: 11000000.10101000.00000000.01000001 → 192.168.0.65/27

Last usable IP: 11000000.10101000.00000000.01011110 → 192.168.0.94/27

Subnet 4-1) Broadcast: 11000000.10101000.00000000.01011111 → 192.168.0.95/27 (Subnet 4-1)

01-11-24
S-4: Performing variable length subnet masking (VLSM):

Subnet 8: Network ID: 192.168.0.224/27

N/W ID: 11000000.10101000.00000000.11100000

No. of usable IP address

Subnet mask: 11111111.11111111.11111111.11111100 → 255.255.255.252

- Here, further breakdown of a subnet is done to create smaller IP network with smaller broadcast domain while efficiently distributing IP address with the least wastage. Since, each of the three WAN links are point to point network that requires only 2 IP addresses, we can determine the no. of host bits needed within an IP address to provide 2 usable IP addresses.

Network ID: 11000000.10101000.00000000.11100000 → 192.168.0.224

Subnet Mask: 11111111.11111111.11111111.11111100 → 255.255.255.252

WAN1 VLSM Subnet 1: 11000000.10101000.00000000.11100000 → 192.168.0.224/30

WAN2 VLSM Subnet 2: 11000000.10101000.00000000.11100100 → 192.168.0.228/30

WAN3 VLSM Subnet 3: 11000000.10101000.00000000.11101000 → 192.168.0.232/30

VLSM Subnet 4: 11000000.10101000.00000000.11101100 → 192.168.0.236/30

VLSM Subnet 5: 11000000.10101000.00000000.11110000 → 192.168.0.240/30

VLSM Subnet 6: 11000000.10101000.00000000.11110100 → 192.168.0.244/30

VLSM Subnet 7: 11000000.10101000.00000000.11111000 → 192.168.0.248/30

VLSM Subnet 8: 11000000.10101000.00000000.11111100 → 192.168.0.252/30

VLSM Subnet 1 (R₁-R₂) WAN1

Network ID: 11000000.10101000.00000000.11100000 → 192.168.0.224/30

1st usable IP: 11000000.10101000.00000000.11100001 → 192.168.0.225/30

Last usable IP: 11000000.10101000.00000000.11100010 → 192.168.0.226/30

Broadcast add: 11000000.10101000.00000000.11100011 → 192.168.0.227/30

VLSM Subnet 2 (R₁-R₃) WAN2

Network ID: 11000000.10101000.00000000.11100100 → 192.168.0.228/30

1st usable IP: 11000000.10101000.00000000.11100101 → 192.168.0.229/30

Last usable IP: 11000000.10101000.00000000.11100110 → 192.168.0.230/30

Broadcast add: 11000000.10101000.00000000.11100111 → 192.168.0.231/30

VLSM Subnet 3 (R₁-R₄) WAN3

Network ID: 11000000.10101000.00000000.11101000 → 192.168.0.232/30

1st usable IP: 11000000.10101000.00000000.11101001 → 192.168.0.233/30

Last usable IP: 11000000.10101000.00000000.11101010 → 192.168.0.234/30

Broadcast add: 11000000.10101000.00000000.11101011 → 192.168.0.235/30

08-11-24

CHAPTER 06

Exploring Network Protocols and Services

IRSHITA CN complete

- A technology, rules, procedure that defines how a sender can package and format a message to be sent across a network to a destination host.
 - Without protocols, for communication on network, devices will not format or address a message properly before placing the message onto the network for delivery. The receiver may misinterpret the message due to lack of formatting or addressing.
 - The transport layer in both the networking model (OSI & TCP/IP) is responsible for assigning the source & destination service port number based on the Application Layer Protocol (ALP). Thus, each application layer protocol is associated with a unique Service Port Number that help devices to deliver a message to the appropriate ALP.
- Service Port - 65535
0-1023: well-known Service Port

⇒ File Protocol:

① FTP (File Transfer Protocol):

- It is a file sharing protocol that operates in a Client-Server model, allowing user to connect to a file server to upload and download files over a network.
- It operates on Service Port 20 (SP 20) to allow data transfer between FTP Client and FTP Server while Service Port 21 (SP 21) is dedicated to control commands & function from FTP Client & FTP Server.

1st Phase -

Client PC — Network — FTP Server

Control Connection

- The computer with FTP Client application open the connection to FTP Server on Service Port Number 21 (SP 21).

2nd Phase -

Client PC — Network — FTP Server

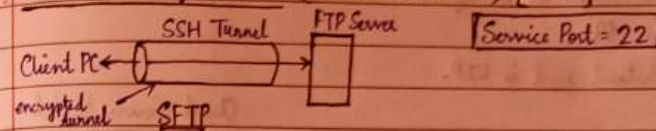
Data Connection

- The client open another connection to the Server on SP 20 to transmit data traffic. (Service Port 20)

3rd Phase - Client PC — Network — FTP Server
Data Transfer

- The data is transferred from the FTP Server from Service Port 20 to the Client.

② SSH File Transfer Protocol (SSH - Secure Shell) [SFTP]



- It allows a client to establish an encrypted tunnel using SSH to a file server that supports the SFTP Protocol.
- Once the SSH connection is established between client & server, both devices encapsulate the FTP packets within the SSH Tunnel for file transfer over the network.
- Using SFTP, it prevents the hacker from identifying any confidential data that is being exchanged over the network.

③ FTPS (File Transfer Protocol Secure):

- It is also referred as FTP over SSL (FTP/S), is a file transfer protocol that allows user to securely transfer file between a client & a server over a network.
- FTPS uses SSL (Secure Socket Layer) or TLS (Transport Layer Security) to encrypt the FTP messages that are being sent over the network between a client & a server.
- It operates on Service Port 990 and Service Port 21.

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→ Service Port 990:

If a client establish a connection to the server on Service Port 990, is considered to be implicit FTPS.

→ Service Port 21:

When the Service Port 21 is used it is called Explicit FTPS.

Explicit FTPS -

When using Explicit FTPS, the client connect to SP 21 on the server and want to SSL with the server, additional steps are taken by sending either AUTH SSL or AUTH TLS command from the client to the server.

AUTH SSL - Authorized SSL

AUTH TLS - Authorized TLS

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This FTPS is called as explicit as the client has the option to enable better security mechanism when required.

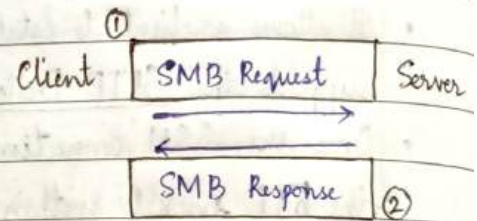
④ TFTP (Trivial File Transfer Protocol):

- Connectionless, light-weight version of FTP.
- Service Port Number - 69
- Transport Layer Protocol used is UDP.

Client-Server Model

⑤ SMB (Server Message Block):

- This protocol operates in Client-Server Model to share network resources in Microsoft Windows Environment.



- Functions:
- i) Starting, authenticating, terminating session between client and server.
 - ii) Controlling access to file and printers.
 - iii) Allowing application to exchange information between devices on a network.

It operates on ^{SP 137}Service Port 137, ^{SP 138}Service Port 138, ^{SP 445}Service Port 445.

SP 137 : TCP/UDP

SP 138 : UDP

SP 445 : TCP

- The client send an SMB Request message to the server to request the shared resources on the server. The server respond with SMB Response message to the client providing requesting additional information for authentication and also provide access to the resources.

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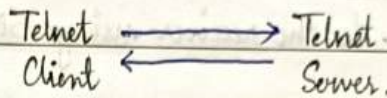
28-11-24 # Remote Access Protocol :

- This protocol allows IT professionals to remotely access & manage device over a network.
- IT professionals configure remote management on their devices, which allows them to remotely connect & implement new configuration or perform troubleshooting to resolve any issue.
- Without RAP, an IT professional will always need physical access to a device to perform any configuration changes or troubleshooting.

- Using RAP provides the convenience for IT professionals to centrally manage end devices, server and networking devices over a network. It also helps to save lots of time and also provide a security risk if IT professional is using an unsecured RAP to a networking device.

① Telnet:

- It is an unsecured RAP that allows IT professional to remotely connect to devices.
- Telnet allows user to access and communicate with a remote device or server using CLI.
- It is a Client-Server application.
- Service Port - 23.



② Secure Shell (SSH): [Service Port-32]

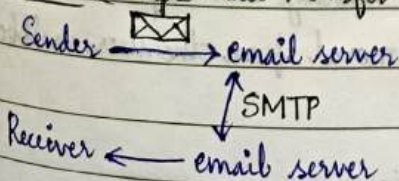
- It is a secured RAP that allows IT professional securely connect to devices over a network to perform configuration changes and troubleshooting.
- Unlike Telnet, SSH encrypt all the messages between the client & the device that is running the SSH service.
- The network analyser i.e., Wireshark, it helps determine the SSH version, encryption algorithm, hashing algorithm for validating the integrity of the message as well as the Message Authentication Code (MAC Value) & the actual encrypted message.

③ Remote Desktop Protocol (RDP):

- It is a secure RAP that is built into microsoft window operating system.
- Using RDP within an organization allows IT professionals to remotely manage window server and desktop devices using GUI (Graphical User Interface).
- Service Port - 3389.
- The encryption takes place by using TLS.

Email Protocols:

① SMTP (Simple Mail Transfer Protocol):



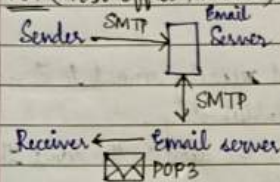
- To ensure emails are transported & delivered over a network, various email protocols help the devices to format the messages for transportation and delivery b/w a sender & destination address.

⇒ The steps to transfer a mail from one end to other end:

- The sender uses an e-mail application such as Microsoft Outlook or Gmail to compose and send the message.
- The email application on the sender's computer uses SMTP to establish a connection to the sender's email server whose Service Port Number is 25.
- When the sender's email server receives the email message, it also uses SMTP to forward the message to the receiver's email server by using SPN-25.
- When the email server receives the mail, the server uses SMTP to send a mail message to the receiver using SMTP.

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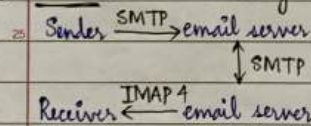
② POP (Post Office Protocol):



- It allows email client to download message over a network.
- The email server that uses POP service listens on TCP Service Port 110.

- Once a TCP connection is made between the client application and the mail server on SP-110, the client downloads the email message from the mail box to the client.
- Once the email has been downloaded, the email messages are deleted from the email server.
- Here, there is no centralised location for storing the message on a network.

③ IMAP (Internet Message Access Protocol):



- This protocol allows email client to synchronize the email message between the client application and email server via Service Port 143.

- When using IMAP on a network, the email messages are kept on the email server until they are manually deleted or removed from the user mailbox on the server.

* SMTP, POP & IMAP are unsecured protocols.

SMTPS

- The secured version of SMTP is SMTPS.
- It uses TLS (Transport Layer Security).
- Service Port Number - 587.

POPS

- The secured version of POP is POPS.
- It uses SSL (Secure Socket Layer).
- Service Port Number 995.

IMAPS

- The secured version of IMAP is IMAPS.
- It uses SSL (Secure Socket Layer).
- Service Port Number 993.

HTTP & HTTPS:

- When communicating with web server / Internet, a user will open a web browser application on their device that uses HTTP to create a message that is recognizable to the web application running on the web server.

→ HTTP: [message is in plain text]

- Unsecured protocol that does not provide confidentiality and data privacy.
- Service Port Number - 80.

→ HTTPS: [message is in encrypted format]

- To resolve the option of security, HTTPS was introduced.
- Service Port Number - 443.
- Security is either provided by SSL/TLS.
- The message that is being transmitted is in encrypted form.

Network Protocol Type (NPT):

- NPT are set of rules that are used to describe how a device communicate with another device over a network.
- If two devices are used, whether they are the same type or different, both system needs to negotiate on a common set of rules. These common rules are referred as NPT.

① ICMP (Internet Control Message Protocol):

- This particular protocol is defined by RFC 792 (Request For Comment 792). is used to provide error reporting on a network.
- Common networking tools like Ping/Trace route are used to invoke ICMP to check end-to-end connectivity between host on a network, identify the path a packet is travelling between a source and a destination & also measure the latency between hops on a network.

Type	Name	Code
0	Echo reply	0
8	Echo request	0
3	Dest ⁿ Unreachable	1 (Host is unreachable) 2 (The Protocol is unreachable) 3 (The Port is unreachable) 4 (Fragmentation needed)
5	Redirect	0 (Redirect from the network) 1 (Redirect from the host)
11	Time Exceeded	0 (TTL exceeded) 1 (Fragment reassembly time exceeded)

② TCP (Transmission Control Protocol): - [Three-Way Handshaking]

- This protocol is defined by RFC 793.
- Connection Oriented Protocol.

③ UDP (User Datagram Protocol):

- This protocol is defined by RFC 768.
- Connectionless Protocol, unreliable, less overhead, faster transmission.

Network Services:① NTP (Network Time Protocol):

- It helps to synchronizing the system clock with the time of Network.
- It allows the IT professionals to configure devices in order to synchronize the system clock/ device clock to the same time on the network.
- It is based on Client-Server model.
- It uses UDP.
- Service Port Number 123.
- Without NTP on a network, IT professional will manually configure the time on their device. This process will be time consuming, which can lead to misconfiguration and mismatch in the time set on other devices.

② DHCP (Dynamic Host Configuration Protocol):

- It provides IP Address automatically.
- It allows network professionals to automatically distribute IP addresses to the client's device on the network.
- When an end-device is connected to a network, it requires an IP address, Subnet mask, Default Gateway and DNS Server (Domain Name System).
- If network professional have to statically/manually assign IP address to each device, the process will be time consuming and leads to duplication of IP address assignment.
- In order to simplify the process, network professional implement DHCP Server, which allows them to configure the following points:-

i) Scope - Range of IP Address in the pool.ii) Exclusion Ranges - IP addresses that should not be distributed on the network.iii) Reservation - Some of the IP addresses are reserved from the pool.iv) Dynamic Assignment - Some of the IP addresses are dynamically assigned to a client on the network.v) Static Assignment - Some IP addresses are statically assigned to the client.vi) Lease Time - The set of time that the client can use the IP address given from the DHCP Server.vii) Scope Option - Additional operation that can be configured when creating the scope.viii) Available Lease - It identify the available lease time for an IP address.

- DHCP provides 4-Way Handshake when the client is connected to the DHCP Server.

DHCP Discover message (BB:BB:BB:BB:BB:BB)
Client → Request → Server

Source MAC	Destination MAC	Source Port	Destination Port
AA:AA:AA:AA:AA:AA	BB:BB:BB:BB:BB:BB	68	67

DHCP Offer message
Client ← Offer ← Server

Source MAC	Destination MAC	Source Port	Destination Port
BB:BB:BB:BB:BB:BB	FF:FF:FF:FF:FF:FF	67	68

DHCP Request message
Client → Request → Server

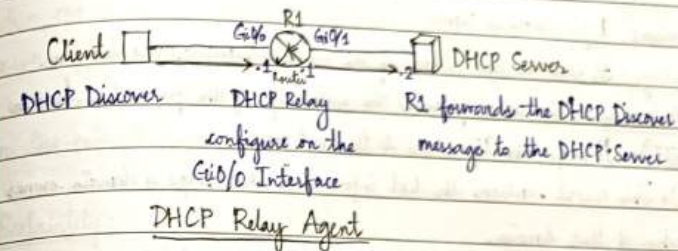
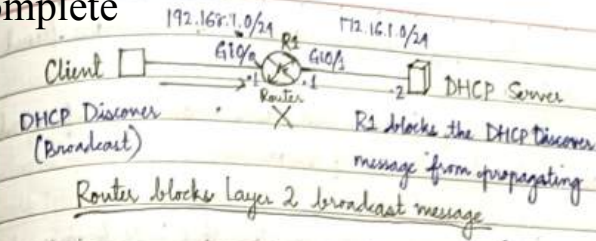
Source MAC	Destination MAC	Source Port	Destination Port
AA:AA:AA:AA:AA:AA	FF:FF:FF:FF:FF:FF	68	67

DHCP Acknowledgement message
Client ← Acknowledgement ← Server

Source MAC	Destination MAC	Source Port	Destination Port
BB:BB:BB:BB:BB:BB	AA:AA:AA:AA:AA:AA	67	68

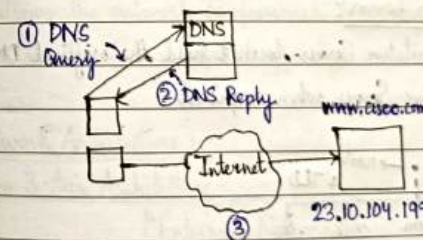
STEPS:

- The client connect to the network and send a DHCP Discover message, seeking a DHCP Server on the network. The client includes the source MAC address and its port number (68), destination MAC address as FF:FF:FF:FF:FF:FF that is the broadcast MAC address for layer-2 network & its port number 67.
- DHCP Server respond with a DHCP Offer message, which contain the IP address needed by the client for communication on the network. This message can be unicast or broadcast.
- The client send a DHCP request message to DHCP Server indicating that it will use the IP address that is available in the offer message. The client send a layer-2 broadcast address to the DHCP Server even though it knows the address from the offer message.
- The DHCP Server respond with DHCP acknowledgement message (unicast) to confirm that the client can use the IP address from the addressing pool of the server.



- Router and other layer 3 devices (that operates in Network layer) don't propagate layer 2 broadcast message beyond their source network. As a result, the DHCP Discover message from the client will not be able to reach the DHCP Server on another network.
- To solve this issue configuring the router to be a DHCP Relay agent will allow the router to forward DHCP message between client and DHCP Server over a network.

③ DNS (Domain Name System): mapping of IP addresses with the host name



⇒ Different DNS Operations:

- The client wants to establish a connection to e.g. www.cisco.com but doesn't know the IP Address of the web server. Therefore, the client sends a DNS Query message to the DNS Server on the Service Port - 53, requesting the IP address of the given website.
- The DNS Server receives the DNS Query message and perform a lookup within its database and find the record. The DNS Server respond with a DNS Reply message that contains the IP address of the given website.

- iii) The client receives the DNS reply message & connect to the IP address found in the response from the DNS Server.
- iv) Within a DNS Server, network professional can create various types of record containing specific IP addressing information. The list of records are:-
- Address.
 - Canonical Name (CNAME): This record provides an alias that is to be mapped with the domain name. Eg., Canonical Name
 - Mail Exchange: It contains the address of the mail exchanger on a domain.
 - Start of Authority: This record provides the authority of the particular domain.
 - Pointer (PTR): It maps the IP address to the host name.
 - Text (TXT): This record contains the text information that helps a domain owner to validate the ownership of that domain.
 - Service (SRV): This record provides the Service Record of the domain.
 - Name Server (NS): It contains the Name Server of the particular domain.

⇒ Two types of DNS Server:

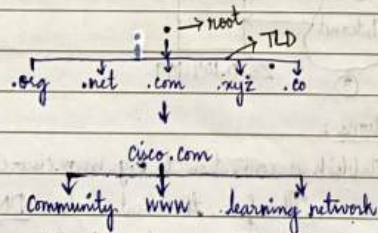
i) Authoritative DNS:

- Final holder of an IP address for a domain name.
- This server contain the original DNS records that are associated with a domain.

ii) Recursive/Non-Authoritative DNS Server:

- Recursive DNS Server/Non Authoritative Server doesn't hold the original DNS record for a domain but ask Authoritative Server when required.

⇒ Domain Name:



- ### # Understanding Network Architecture: It is essential to understand the importance of designing an optimal network architecture to support both current & future demand of an organization.
- Many organization experience congestion, high latency (slow response). Thus, while designing a network following factors are to be considered:
- Fault Tolerance & Redundancy:
Fault tolerance is the ability to continue functioning and providing its service to a network, when one or more components are affected. Network must be designed with fault tolerance to ensure if a networking device has a failure, the network quickly detect the failure and recover to forward the information between the source & the destination. The fault tolerance on the network is been created by redundancy of multiple path between source & the destination.

ii) Scalability:

It allows a network to easily grow in order to support new devices, application and services without affecting the performance of the existing services.

iii) Security:

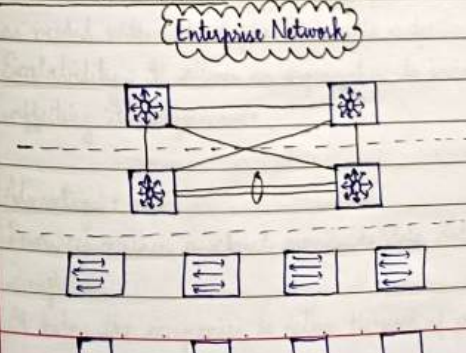
With the increase in cyber attack, organization needs to safeguard their assets from the hacker. One of the security solution is firewall appliances.

iv) Quality of Services (QoS):

It allows the network professionals to configure networking devices to prioritize network bandwidth for specific traffic type.

Network Architecture:

i) Cisco 3-tier Architecture:



• Access layer-

- It is also called as edge layer.
- It is responsible for providing network connectivity and access from end-devices to the network and resources. In this layer, there is no redundancy for end-devices that are connected to the network.

• Distribution layer-

- It is also called as Aggregation layer.
- This layer provides link aggregation and redundancy for the access layer switches on NW.
- There are two distribution layer routers within the Branch office, each access layer switch will connect to each distribution layer routers to ensure redundancy between the end devices.

• Core layer-

- This layer is the high speed backbone of the network architecture.
- It provides interconnectivity and redundancy for the distribution layer within organization with many branch offices.

Advantages: Scalability, Redundancy, Fault Tolerance

Disadvantage: Expensive.

ii) Cisco 2-tier Architecture:

- It is designed for smaller organization that consists of small network and small budget. The collapsed core layer is a fusion of distribution layer and core layer. Either of the router can be used in collapsed core layer. Access layer provides the same function as that of the 3-tier Architecture.

Fundamental of Cloud Computing:

- Cloud computing providers use a model called pay-as-you-go model.
- Cloud - virtual storage.

Advantage: • Saves the cost of hardware storage & physical existence.
• Anywhere you can use the service.

Features of Cloud Computing:

- Storage
- Grows the cloud infrastructure and scale down.
- High up time
- Fault tolerance
- Elasticity and Scalability increases.

- The word 'cloud' means the customer is using a service application or server within a data center that cannot be seen by the customer or physically accessible.
- Cloud providers allow their customers to quickly spin up (create) a new virtual server on the cloud. While allowing the customer to scale up or scale down the resources on the server. Scaling up is possible by adding features. Scaling down is possible by reducing resources.

- Tenant is a single customer on a cloud provider platform.

- Multi Tenant - When multiple customer uses the same cloud provider platform.

- Tenant Isolation - It allows multiple customers to use the same shared resources within the data center but each customer is isolated from accessing another customer's resources.

• Cloud computing provides supports - elasticity & scalability.

- Elasticity: It allows a system to adapt to the workload and changes of the environment such that a customer can quickly provision and de-provision servers and applications as needed within cloud provider's data center.

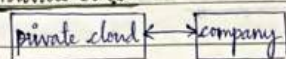
- Scalability: It allows an organization to increase or decrease the network without affecting its performance.

Advantages:

- Resources within a cloud provider's data center can be accessible from anywhere at anytime.
- It helps the companies to reduce the need of physical existence of the resources.

Cloud Deployment Model:

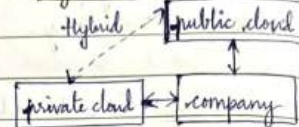
① Private Cloud:



② Public Cloud



③ Hybrid Cloud



④ Community Cloud

① Private Cloud:

- This model is being owned by an organization where the resources are only used by the employee of that organization.

② Public Cloud:

- This model is available to anyone on the Internet. It allows anyone to create virtual server & deploy applications and services on a data center that share its resources with other and uses a 'pay-as-you-go' model.

③ Hybrid Cloud:

- It is a mixture of Private & Public Cloud Deployment Model.

Important Characteristics -

Main Advantage

- It allows an organization's data, server & applications to be locally backed up on its private data center and replicated to an online public cloud solution provider.

④ Community Cloud:

- It is where several organization share the resources on a cloud platform. These can be group of companies with similar interest or partnership will share the resources within a data center or a cloud service provider.

Cloud Service Model:

① SaaS (Software as a Service):

- This service model allows the cloud solution provider to offer access to the user interface of the application that is being hosted within the cloud provider's data center. The user is neither concerned nor has access to manage the host operating system that are hosting the application.

② PaaS (Platform as a Service): This service model is designed for developer who require a bit more control over the operating or the working environment.

③ IaaS (Infrastructure as a Service): It allows the user to manage physical hardware and software resources on the virtual server on the cloud provider's data center.

④ DaaS (Desktop as a Service): Here the Cloud Service Provider can deliver a virtual desktop environment to a user over the Internet. The Cloud Service Provider is responsible for managing all the backend maintenance. DaaS provides two offer: i) Persistent Desktop

ii) Non-Persistent Desktop.

i) Persistent Desktop: It keeps the data and the changes made by the user.

ii) Non-Persistent Desktop: Each time the user log off from the virtual desktop, all data & settings are cleared from the system.

⑤ IaC (Infrastructure as Code): This model focus on managing the resources within a Cloud Service Provider's data center and the virtual machines.

Cloud Connectivity Solution:

i) Using a VPN which allows a secure, encrypted connections which is been established over a unsecured network i.e., Internet. Using a VPN allows the company to save lots of money while protecting data in motion over the Internet.

ii) Using a Private Direct Connection to the Cloud Provider.

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CHAPTER 04 Routing & Switching Concepts

IRSHITA CN complete

Ch 4 onwards refer your book.

Exploring Routing Concepts:

- Routers are Layer-3 devices that operates at the Network Layer of OSI and Internet Layer of TCP/IP.

⇒ Functions of Router:

- They interconnect two or more different networks. Forward the packets to the destination.
- They provide NAT operation & filtering traffic using Access Control List.
- Routing algorithms are divided into two types:

a) Dynamic Routing b) Static Routing.

- In case of Dynamic Routing, the router calculate the best path from the source to the destination by assigning metric values to the various factors like-

- Cumulative Bandwidth RIP- Routing Information Protocol.
- Latency EIGRP- Enhanced Interior Gateway Routing Protocol.
- Reliability OSPF- Open Shortest Path First
- Outgoing Load (Transmitted load). IS-IS- Intermediate System-Intermediate System
- Receiving Load
- Hop Count
- Path

- A router will check its routing table using a Top-Down Approach until it finds the suitable path. Once the suitable path is found, the router forward the packets to the destination.

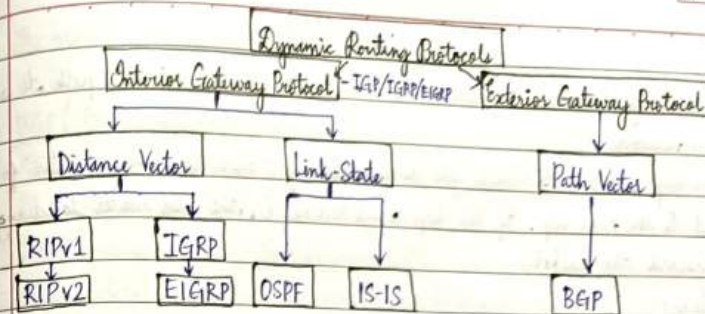
- A network route specify ① exit interface of the router.
② next hop address.

a) Dynamic Routing:

This protocol are designed to help routers automatically learn about designation N/W by sharing routing information with other routers.

Advantages:

- It provides best path to forward packets to the destination.
- It can find new best path if current path isn't available.
- Maintain an up-to-date routing table on all routers within the N/W.
- It finds out remote network by sharing routing information between routers on a N/W.



- Distance Vector - This protocol forward packets based on distance & direction.

Distance Vector Algorithms consists of following factors:

- It exchange routing information between neighbour routers that are configured within the same routing protocol.
- It send its entire routing table to immediate neighbour router.
- It contains a mechanism for calculating best path to the destination and add the route in the routing table.
- It can detect and adapt to changes within the network topology & update the routing table.

i) RIP: Routing Information Protocol

EIGRP: Enhanced Interior Gateway Routing Protocol.

- Link-State Protocol: It uses cumulative bandwidth to the destination network. The routers that are configured with this protocol build its topological map of the entire network, which helps the router to determine the shortest path to the destination. It will only send the update to the neighbour router if there is a change in the topology.

e.g., OSPF- Open Shortest Path First

IS-IS- Intermediate System-Intermediate System.

i) RIP (Routing Information Protocol)-

- Uses an algorithm → Bellman-Ford algorithm and finds the best path.
- Distance is measured in terms of no. of hops.
- Maximum Number of hop = 15.

- Bellman-Ford algorithm uses hop count as a metric to calculate the best path to a destination network.
- Maximum hop count is 15 which gets decremented by 1 each time when router forwards the packet to the next hop. If the hop count reaches 0, the last router to change it to 0 will discard the packet.

Disadvantage:

- It doesn't support a larger network with hop count greater than 15.
- It broadcasts the entire routing of each router every 30 seconds, regardless of whether a network topology change occurred or not.
- RIPv1 doesn't support networks that use custom subnet mask.
- RIPv2 supports VLSM but doesn't manually specify the custom subnet masks.

ii) OSPF (Open Shortest Path First):

- It's a Link-State Dynamic Routing Protocol that uses its own datagram and is tagged in IP protocol as Protocol Number - 89.
- It doesn't use TCP or UDP.
- It uses Shortest Path First algorithm which determine the cumulative bandwidth to a destination network and choose the path with low-cost.
- Being a Link-State Dynamic Routing protocol, each OSPF enabled router exchanges Hello packets with their neighbour router in order to provide mutual handshake.
- Once the handshaking is over, each router will exchange Link State Advertisement (LSA) information i.e., the state and the cost of the link between the neighbour router.
- Once the routers collect the LSAs from each router, the Link State Data Base (LSDB) i.e., topology table is created.
- This topological table is been shared among all the routers.
- After that SPF algorithm is used to determine the shortest path.

iii) EIGRP (Enhanced Interior Gateway Routing Protocol):

- This protocol uses Protocol Number - 88 which is encapsulated directly into IP datagram rather than TCP or UDP.
- It uses Diffusing Update Algorithm to find out the best path to the destination and Backup Path also to the destination.

- The metric that is used is Bandwidth & Delay.

iv) BGP (Border Gateway Protocol):

- It is a Path Vector Routing Protocol that is used by ISPs/ASs (Autonomous Systems) to exchange routing information with each other.
- BGP operates on TCP - Port Number - 179.
- This protocol is very slow converging routing protocol that doesn't send updates as soon as a change occur on the network topology but wait for a while for the network to be restored.

BGP is divided into two parts -

1) eBGP (External BGP)

- used between different ASs.

2) iBGP (Internal BGP)

- used between a single AS.

b) Static Routing:

Different types of Static Route:

- Standard Route: This route specifies how to reach a destination network.
- Default Route: When no other route matches the destination address of a packet.
- Floating Route: It's a backup route to a primary route on a router.
- Summary Route: It represent multiple destination IP network in the form of a Single Consolidated Static Route.

Bandwidth Management: (Refer your textbook)

Network professional uses two types of solution to ensure specific traffic types have a high priority of accessing the bandwidth over others on a network. They are:

- Traffic Shaping
 - Quality of Service (QoS)
- i) Traffic Shaping allows network professionals to create delay of some traffic types using a traffic profile. By doing so, latency is improved and optimize the performance of a network while increasing the allocation of bandwidth for specific traffic type.
- ii) Quality of Service (QoS): The metric that are used within QoS to measure type of traffic are Bandwidth, Delay, Congestion, Packet Loss, Jitter (measures the variation in the delay time of incoming packets on the network.)
- QoS control traffic on a network while ensuring the improved performance.

Steps for prioritizing the traffic using QoS:

- Classification:** Inbound traffic on networking devices is inspected & placed within a waiting queue based on the DS field (Differentiated Service field) of IPv4 packet.
- Marking:** This is the process where QoS tool modify one or more field within the packet header to insert a value.
- Queuing:** It manages all the queues for outgoing messages on the device.
- Policing & Shaping:** Policing discard & drop packet whereas Shaping keep/hold the packet within the queue.
- Congestion avoidance:** It reduces the amount of congestion.

Diving into Switching Concepts:

→ Spanning Tree:

- When designing a network topology, it is essential to implement redundancy and fault tolerance to ensure multiple path that are available between the source & destination.
- When multiple switches are interconnected to create redundancy, Layer 2 loop and duplicate messages are been created.
- To resolve this issue, Spanning Tree Protocol (STP) was designed which can detect any physical Layer-2 loop on a network and logically block the redundant path i.e., only one active logical path exists between the source and the destination.
- When the switch is on, it send Bridge Protocol Data Unit frame (BPDU) every 2 seconds to its neighbour switch.

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Spanning Tree Protocol (STP):

Step 1: Identifying the root bridge (central referenced point for all the traffic being forwarded to LAN).

Step 2: Identifying the root ports (interface)

The ports closest to the root bridge but not in the root bridge.

Step 3: Identifying the designated port.

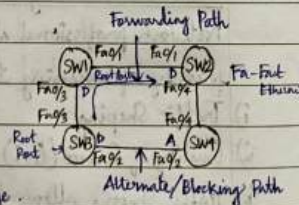
Any non-root port in a forwarding state.

Step 4: Identifying the alternate/blocking state.

Any port that are logically blocked by STP to prevent the Layer-2 loop due to redundant paths.

* The switch which is having the lowest priority value among all is considered to be ROOT BRIDGE.

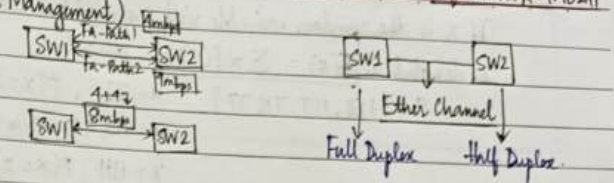
* If the priority value is same of all switches, then the switch having the lowest MAC ID is considered as the Root Bridge.



⇒ Port Aggregation: (Bandwidth Management)

Conditions:

- Same type of interface.
- Same number of interface.
- Same configuration.
- Same speed.

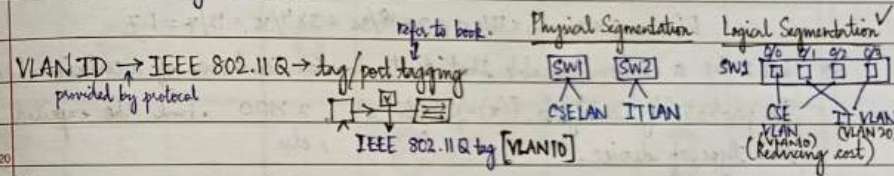


⇒ Neighbor Discovery Protocol:

- **CDP:** Cisco Discovery Protocol (Finds the next hop)
- **LLDP:** Link-Layer Discovery Protocol. [When all devices are Cisco & non-Cisco, then this is used].

⇒ VLAN (Virtual Local Area Network):

- Benefits:**
 - Improving N/W performance & bandwidth.
 - Reducing the size of broadcast domain.
 - Improving network security.
 - Reducing the cost.



There are 2 types of links/interfaces:-

i) **Access Port:** It is used to connect PC to Switch.



ii) **Trunk Links:** It is used to connect Switch to Switch, Router to Router, Switch to Router.

→ **Access Port on Switch** is used to connect PC to switch. It allows only one statically assigned VLAN on the interface.

→ **Trunk Links** are point to point connections from S to S, R to R, S to R which allows multiple VLANs to carry the traffic between switches and routers.

Chapter - 09

⇒ Types of VLAN:

i) Default VLAN:

- It is created by the vendor.
- By default, the interface of switches are assigned to VLAN 1.
- It is not recommended to use Default VLAN for security reasons.
- It is not possible to rename or delete the default VLAN on a switch.

ii) Data VLAN:

- It is configured to transport traffic generated by 'n' devices such as computers, printers, servers and access points.

iii) Native VLAN:

- It is used to transport untagged traffic on IEEE 802.1 Q Trunk Link.
- Untagged traffic doesn't originate from a VLAN. It is a self-generated traffic from the switch itself such as CDP & LLDP messages.
- The receiving switch ports inserts an IEEE 802.1 Q Tag, VLAN ID into the frame. This traffic is known as Tagged Traffic.

iv) Management VLAN:

- It is used to remotely access the switch over a network for management purpose.
- It is referred to as a switch virtual interface and is configured with an IP address and a subnet mask.

v) Voice VLAN:

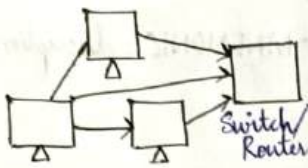
- Voice traffic uses UDP.
- Since a coverage network is recommended type of network infrastructure, having a dedicated network for all voice traffic is preferable.
- Using a dedicated VLAN to transport voice traffic will ensure that all the voice traffic is kept separated from the other traffic types on the physical network.

Chapter 10 : Wireless Networking

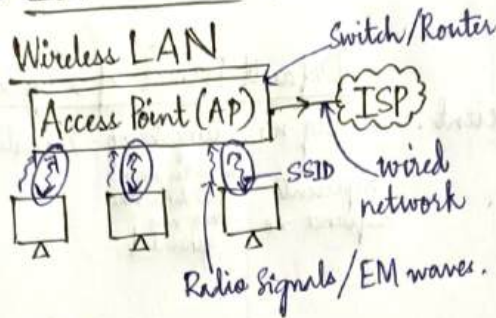
IRSHITA CN complete

- ⇒ WLAN → connected wirelessly
→ communicate through Radio signals/Electromagnetic waves.
- ⇒ WLAN follows a protocol called IEEE 802.11 (Wi-Fi).

Wired LAN



Wireless LAN



21-12-24.

Refer to pg 406 & 407 : For Wireless Router.

- ⇒ SSID (Set Service Identifier): These are the name of wireless routers that differentiate from other wireless routers.
- ⇒ Beacons: These are the broadcast messages/broadcast frames that are transferred periodically by wireless n/w to know the client's existence.
 - Beacons transmit - SSID, Encryption, MAC.

⇒ Station: Clients

- ⇒ Probes: Every station probe into the Preferred Network List (PNL).
 - It is a tool that monitors the activity/success of a network.

Refer to Pg 408 & 409: For SSID, Beacons, Station, Probes definition.

Chapter 11 : Assuring Network Availability

• Performance metrics help:

- to determine whether the network is operating as expected.
- to know the delivery of network resources is affected.

• Common performance metrics used:

- Temperature
- CPU Utilization
- Memory
- Jitter: measures the variation of delay times of incoming packets on a n/w
- Latency: time b/w a request & a response over a network.
- Bandwidth: Network bandwidth is simply the total amount of packets that can be transferred from a source device to destination device in a given time.

⇒ Simple Network Management Protocol (SNMP):

- It is a common network protocol, allows network professionals to easily communicate/monitor device
- It has three versions: i) SNMPv1
ii) SNMPv2
iii) SNMPv3

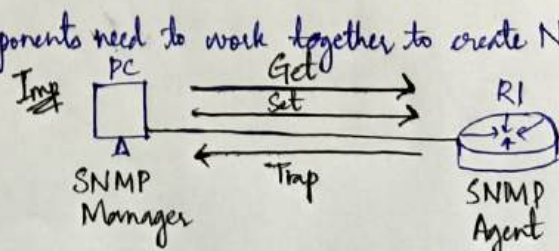
⇒ Network Management System (NMS):

When working with SNMP, three main components need to work together to create NMS.

1. SNMP Manager

2. SNMP Agent

3. MIB (Management Info Based)



→ Network device logs:

- Log messages contain timestamps, severity levels and description of events displayed.
- Traffic log
- Audit log

Syslog:

- Very time-consuming & inefficient.
- Centralised logging server.

Default Format of a Syslog Message:

seq no: timestamp: % facility-server % - MNEMONIC: description
 ↓
 represents sequence no. includes date and time the msg was generated

- Syslog Protocol uses: UDP
- Service Port Number: 514

Chapter 12: Organizational Documents and Policies

→ Change Management Board

→ Phases of Change management:

- Request
- Evaluate
- Authorize
- Implement
- Documentation.

⇒ Incident Response Plans:

- Performed by Cyber Incident Recovery Team (CIRT), commonly used by cyber-security team.

Preparation



Detection & Analysis

- event
- incident

- Steps:
- Preparation
 - Detection & Analysis
 - Containment, eradication and recovery.

→ Post-incident analysis

⇒ Business Continuity Plan (BCP)

Consider following factors:

- Exercises (tabletop)
- After-action reports
- Failover
- Alternate Disaster Practices.

→ Disaster Recovery Plans: